

The near-gap protection lemma: operator monotonicity closes the G-U1-SMALLT regime structurally

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-06 · this version issued 2026-06-06 · v1.0

1. Purpose and scope

Unit U9 (second-order audit of U1) registered G-U1-SMALLT: the U1 composition's coverage of near-gap small-amplitude competitors at the certified intensities was unverified, with route (a) (structural protection) and route (b) (crossover arithmetic) as candidate resolutions. This note executes route (a). Scope: the chain's stated convention (B5 main note v2.0, Setting): $D(k) = \hat{r} + C(k^2 - q_0^2)^2$, $\hat{r} = r_R + 2\lambda'I$, $r_R = 0.3045257$ fixed, $\lambda' = 8.0551 > 0$ certified, $W = \lambda'(P^2 - 2I \text{Id})$ (the P^2 -representation theorem, main chain v1.7–v1.8).

2. The lemma

Lemma 1 (Near-gap protection – sketch). *For every admissible pattern $\mathcal{P}$ of intensity I , within the chain's stated convention, the competitor's full fluctuation Hessian obeys*

$$H_{\mathcal{P}} = D + W = D_0 + \lambda'P^2 \geq D_0 = H_{\mathcal{R}_H}, \quad D_0 = r_R + C(k^2 - q_0^2)^2,$$

and therefore, by the operator monotonicity of $\ln$ on positive operators,

$$F_{\text{fluct}}[\mathcal{P}] - F_{\text{fluct}}[\mathcal{R}_H] = \frac{1}{2V} \text{tr}[\ln(D_0 + \lambda'P^2) - \ln D_0] \geq 0.$$

The fluctuation sector — diagonal Hartree shift ($+2\lambda'I$) and off-diagonal Bloch channel (W 's off-diagonal part) TOGETHER — can never lower a competitor below $\mathcal{R}_H$.

Proof sketch. $D = D_0 + 2\lambda'I \text{Id}$ by the convention; $W = \lambda'(P^2 - 2I \text{Id})$ by the P^2 theorem; adding gives the exact identity $D + W = D_0 + \lambda'P^2$ (this is the chain's own display, v1.7). P is self-adjoint (multiplication by the real field $F(x)$, v1.8), so $P^2 \geq 0$; $\lambda' > 0$ is certified at the anchor band. Hence $H_{\mathcal{P}} \geq D_0 > 0$. Operator monotonicity of $\ln$ ($A \geq B > 0 \Rightarrow \ln A \geq \ln B$, Loewner) gives $\text{tr} \ln H_{\mathcal{P}} \geq \text{tr} \ln D_0$, and the Gaussian fluctuation free energy is $+\frac{1}{2V} \text{tr} \ln(\text{Hessian})$. $\square$

3. Consequence: G-U1-SMALLT closes at the regime level

The U9 crossover ($O(I)$ floor vs $O(I^2)$ envelope) compared two BOUNDS; the lemma shows the underlying ENERGIES cannot cross in the fluctuation sector at all. The competitor's total matched excess decomposes as

$$\Delta F[\mathcal{P}] = \underbrace{\Phi(t)}_{\geq 0 \text{ on } I'} + \underbrace{\Delta F_0\text{-type layer excursion}}_{\geq 0} + \underbrace{F_{\text{fluct}}[\mathcal{P}] - F_{\text{fluct}}[\mathcal{R}_H]}_{\geq 0 \text{ (this lemma)}}$$

so on interval I' (all Φ coefficients positive; contains the gap point) EVERY term is nonnegative and the selection cannot flip — no margin needed. The role of the STEP-5B bound is thereby clarified to exactly the chain’s original threat model: on the BAND/sub-band intervals, where the layer proof spends the P_B floors against the dips, the off-diagonal correction must not consume the residual $+0.00432$ — and there the U1 composed arithmetic stands as written. The U9 estimated crossover inside the certified window is thus an artifact of bounding the protected sector instead of using its positivity.

4. Named residuals (honest)

- **R-U10-1 (convention exactness):** $\hat{r} = r_R + 2\lambda'I$ with FIXED r_R treats the pattern’s Hartree shift linearly; the competitor’s self-consistent $M(\hat{r})$ feeds back into λ' and the sub-leading dressing at $O(I^2)$. One page: show the feedback is either part of the matched bookkeeping’s definition (in which case the identity is exact by convention) or bounded by the already-certified λ' -calibration band (the artefact’s lam_calibration block measures exactly this at the LAM point: $\delta_{\text{second order}} = 6.7 \times 10^{-8}$).
- **R-U10-2 (split alignment):** the lemma’s inequality is for the TOTAL fluctuation trace; the chain’s layer/beyond-layer split books the diagonal $2\lambda'I$ shift inside the layer’s $F_0(\hat{r})$ -excursion and the off-diagonal inside δF_{off} . One page: verify no piece of the trace is double-counted or dropped when the split is reassembled (the v1.7 identity is the algebra; the page is bookkeeping).
- **R-U10-3 (machine check, registered):** numerically verify $\text{tr}[\ln(D_0 + \lambda'P^2) - \ln D_0] \geq 0$ on test patterns and that its small- I expansion reproduces (diagonal gain) – (off-diagonal w^2J form) term-by-term — one small script (not executable this session; sandbox shell unavailable).

Upon R-U10-1/R-U10-2 (two pages) the U9 flag lifts and the U1 proposal returns to “operator sign-off required” only.

5. Numerical verification

No new machine numbers. The lemma is structural; its inputs ($\lambda' = 8.0551 > 0$; the P^2 identity; r_R) are certified in the existing artefact (192/192). The small- I consistency target for R-U10-3: first order of the trace $= \lambda' \langle F^2 \rangle M(r_R)$ -form $= 2\lambda'IM_R$ -scale $\approx 1.76I$ — linear and positive, dominating the $O(I^2)$ off-diagonal at all certified intensities (at $I = 2 \times 10^{-3}$: 3.5×10^{-3} vs the 1.7×10^{-3} envelope), consistent with the positivity.

Quantitative sanity check (mandatory): limit case — at $I \rightarrow 0$, $P \rightarrow 0$ and the inequality degenerates to equality (the competitor IS $\mathcal{R}_H$), as it must; sign-direction — $\lambda' > 0$ is the load-bearing sign (“pattern Hartree RAISES the diagonal,” a certified artefact claim with its own assert); had $\lambda' < 0$ the lemma would fail and the U9 flag would stand — the lemma’s validity is pinned to a machine-asserted sign, not an assumption; consistency — the first-order trace coefficient ≈ 1.76 per unit I exceeds the U9 floor coefficient $m^* = 0.3045$ per unit I : the structural protection is STRONGER than the layer floor the U9 audit was comparing against, explaining why the bound-level crossover was an artifact.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The lemma proves the fluctuation sector never helps competitors — but then why does the chain bound $|\delta F_{\text{off}}|$ at all? If the total is positive, STEP-5B was unnecessary.” DISMISSED: the two statements protect DIFFERENT regimes. On the band intervals the layer proof itself goes NEGATIVE in places (the dips) and survives by the P_B floors — there the decomposition books the diagonal Hartree gain inside the layer excursion, and the residual question is whether the off-diagonal piece (which is genuinely sign-indefinite within the split) can eat the remaining +0.00432; that is STEP-5B and it was needed. The lemma adds the complementary statement for the regime where the floors vanish. Both are necessary; neither subsumes the other.
- (β) “Operator monotonicity needs $D_0 + \lambda' P^2$ and D_0 to act on the same space with the trace difference well-defined (the operators are unbounded; $\text{tr} \ln$ is formal).” VALID-with-mitigation: per unit volume the difference of the two $\text{tr} \ln$ ’s is the standard regularized log-determinant ratio, finite because $\lambda' P^2$ is bounded (multiplication by $\lambda'(F^2)$, $\|F\|_\infty^2 \leq 4S^2$ finite for finite patterns) and relatively trace-class against D_0^{-1} decay (k^{-4}); the chain’s own Lemma B manipulates exactly this object. The one-paragraph regularization statement folds into R-U10-2.
- (γ) “R-U10-1 could be where the problem hides: if the r_R -fixed convention is only first-order, the identity $D + W = D_0 + \lambda' P^2$ with $D_0 = H_{R_H}$ is approximate and the positivity could fail at $O(I^2)$ — the very order of the threat.” VALID-with-mitigation — this is exactly why R-U10-1 is named rather than waved through: the artefact’s own second-order calibration ($\delta_{\text{second order}} = 6.7 \times 10^{-8}$ at the LAM point, versus the protected linear gain $\sim 1.76I = 1.8 \times 10^{-4}$ there) measures the feedback at $\times 2600$ below the protection at the calibration point, strongly indicating the convention’s slack is harmless; the one-page writeup must make that class-wide. Until then the lemma is T3 and the U9 flag stands in REDUCED form (two pages, not an open regime).

7. Result footer

Result ID:	B1-RH-ENUM / neargap-protection-lemma (unit U10; closes U9’s G-U1-SMALLT route (a))
Precise statement:	LEMMA (T3 sketch): $D + W = D_0 + \lambda' P^2 \geq D_0 = H_{\{R_H\}}$ (chain identity + $\lambda' > 0$ certified) $\Rightarrow$ by operator monotonicity of $\ln$, $F_{\text{fluct}}[P] - F_{\text{fluct}}[R_H] \geq 0$ for every admissible pattern: the fluctuation sector never helps competitors. With $\Phi \geq 0$ on interval I' , the near-gap small-amplitude regime is protected structurally; the U9 crossover was a bound-level artifact. G-U1-SMALLT reduced to R-U10-1 (convention exactness, one page) + R-U10-2 (split alignment, one page) + R-U10-3 (machine check, registered).
Scope:	The chain’s stated convention (v2.0 Setting); all certified intensities; amended class.
Dependencies:	P^2 -representation theorem (main chain

	v1.7-v1.8); $\text{lam}' > 0$ (artefact assert); U9 audit (the item being closed); Prop A interval I' structure.
Evidence grade:	ANALYTIC SKETCH (T3; core step = textbook Loewner monotonicity).
Reproduction command:	(registered R-U10-3) small script verifying the trace inequality + small-I expansion on test patterns.
Expected output:	trace difference ≥ 0 on all test patterns; first-order coefficient $\sim 2 \text{ lam}' M_R$; JSON under claims/B1-RH-ENUM/runs/.
Falsification gate:	$\text{lam}' \leq 0$ anywhere on the certified band (contradicts a certified assert); an $O(I^2)$ convention-feedback term exceeding the linear protection class-wide (would void R-U10-1 and restore the U9 flag in full).
Tier before / after:	No tier action. B1 stays T5. The U1 proposal condition G-U1-SMALLT is REDUCED (not yet lifted): operator sign-off still required, now with two one-page residuals in queue.
No-overclaim statement:	The lemma is a T3 sketch; the regularization and convention-exactness pages are named, not assumed. The U9 flag is reduced, NOT removed. PDF build + linter PENDING operator-side.
Next required action:	R-U10-1 + R-U10-2 writeups (one page each); R-U10-3 script; then the U1 proposal returns to operator-decision-only. Mainline continues (U11).